

Seunghoon Choi

Computational Materials Scientist

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RESEARCH INTERESTS

- Computational materials design using first-principles calculations and machine learning
- Multi-fidelity modeling strategies for accelerated materials discovery
- Structure–processing–property relationships in complex energy materials systems
- Descriptor development linking atomic-scale features to functional performance
- Data-driven optimization of microstructure formation and interfacial phenomena

EDUCATION

Ph.D., Materials Science & Engineering (Computational Materials)

Fall 2026 (incoming)

University of Florida, Gainesville, FL, USA — Advisor: Prof. Yijia Gu. Full graduate assistantship.

M.E., Nano Science and Technology

Mar 2017 – Aug 2019

Sungkyunkwan University, Suwon, Korea — Advisor: Prof. Changhyun Pang.

Thesis: *Conductive Hierarchical Hairy Fibers for Highly Sensitive and Multiple Gesture-Distinguishable Multimodal Sensor.*

B.S., Advanced Materials Science and Engineering

Mar 2010 – Feb 2017

Sungkyunkwan University, Suwon, Korea.

RESEARCH EXPERIENCE

Visiting Researcher — Soft Electronics & Robotics Lab

Jun 2025 – present

Korea National University of Transportation, Chungju, Korea — Host: Prof. Da Wan Kim

- Microstructured polymer interfaces for molecular transport in microneedle-based interstitial-fluid extraction.
- Surface engineering & contact mechanics of microstructured polymer–tissue interfaces for soft-material signal acquisition.
- Grain-boundary engineering in Cu/Sn-doped telluride thermoelectric alloys — composition–property relationships and charge transport.

Graduate Researcher — Polymer 3D-Architecture Lab

Mar 2017 – Aug 2019

Sungkyunkwan University, Suwon, Korea — Advisor: Prof. Changhyun Pang

- Hierarchical percolation networks in stretchable conductive polymer composites (microcrack-assisted dual-resistance transport) → *Adv. Funct. Mater.* 2019, Back Cover.
- Bio-inspired polymer surface engineering for reversible adhesion on wet and curved substrates → MRS Best Poster 2018.

PROFESSIONAL EXPERIENCE

SK On Co., Ltd.

Jul 2021 – Dec 2024

Expatriate Process Engineer — Yancheng, China (Jan 2023 – Dec 2024)

- Commissioned and stabilized industrial-scale lithium-ion battery electrode manufacturing lines during new-plant start-up; led ramp-up troubleshooting across cross-functional engineering teams; validated equipment via SAT/FAT.

Mixing Process Engineer — Republic of Korea (Jul 2021 – Jan 2023)

- Supervised industrial-scale slurry mixing for cathode and anode electrodes; analyzed how mixing variables govern slurry rheology, dispersion stability, and downstream coating performance; applied statistical process analysis to identify critical parameters.

PUBLICATIONS

1. **Conductive hierarchical hairy fibers for highly sensitive, stretchable, and water-resistant multimodal gesture-distinguishable sensor, VR applications.** S. Choi, K. Yoon, S. Lee, H. J. Lee, J. Lee, D. W. Kim, M.-S. Kim, T. Lee, C. Pang. *Advanced Functional Materials* 29(50), 1905808 (2019). **First author · Back Cover · 140+ citations.** DOI: 10.1002/adfm.201905808
2. **Design and Sensing Frameworks of Soft Octopus-Inspired Grippers Toward Artificial Intelligence.** S. Choi, J. Jang, J. Lee, D. W. Kim. *Biomimetics* 10(12), 813 (2025). **Co-first author (invited review).** DOI: 10.3390/biomimetics10120813

3. **Octopus-Inspired Adhesive Electrode for Robust EMG-Based Robotic Control under Dry and Wet Conditions.** S. Jeon, [S. Choi](#), J. Jang, J. Y. Shim, D. W. Kim. *International Journal of Internet, Broadcasting and Communication* 18(1), 274–284 (2026). **Co-first author.** DOI: 10.7236/IJIBC.2026.18.1.274
4. **Bio-Inspired Hydrogel Adhesive Electrode Enabling Stimulus-Responsive Electrical Signal Recording in Plant Leaves.** S. Jeon, [S. Choi](#), D. W. Kim. *International Journal of Advanced Smart Convergence* 15(1), 241–253 (2026). **Co-first author.** DOI: 10.7236/IJASC.2026.15.1.241

Under review: Pilot clinical study of a biomimetic suction patch for enhanced topical delivery. [S. Choi](#) et al. *Journal of Cosmetic Dermatology*, 2026 (2nd revision).

PATENTS

Fiber-Type Sensor and Its Manufacturing Method

KR 10-2026011 · Registered 2019

Functional Surgical Suture with Micro-structured Surface and Its Manufacturing Method

KR 10-2019-0061944 · Filed 2019

SELECTED PRESENTATIONS

- **Octopus-Inspired Adhesive Patch for Bio-Signal Monitoring.** Materials Research Society (MRS) Fall Meeting, Boston, MA, Nov 2018.
- **A Fiber-Based Multimodal Sensor Using Hybrid Nanocomposites for Human Motion Monitoring.** Polymer Society of Korea Fall Meeting, Busan, Korea, Nov 2017.

AWARDS & HONORS

- **Government Scholarship, Integrated B.S.–M.S. Program** — Sungkyunkwan University, 2017. Full merit scholarship, including an 8-week research residency at ESPCI Paris.
- **Best Poster Award** — Materials Research Society (MRS) Fall Meeting, Boston, 2018.
- **Back Cover Article** — *Advanced Functional Materials* 29(50), 2019.
- **Invited Contribution** — Wiley MaterialsViews China, 2019.

TECHNICAL SKILLS

Characterization & Structural Analysis — SEM (EDS), AFM, Optical Microscopy, XRD, FT-IR, UV-Vis Spectroscopy, Contact Angle.

Processing & Microstructure Fabrication — Microstructure Imprinting, Polymer Surface Patterning, Stretchable Composite Fabrication, Surface Coating.

Slurry & Colloidal Systems — Slurry Rheology, Colloidal Dispersion Stability, Particle Suspension, Industrial Mixing.

Modeling & Data Analysis — COMSOL Multiphysics (FEM), MATLAB, Minitab (SPC), OriginPro, Excel VBA.

Programming & Systems — Python (PyTorch), C++, Statistical Process Control, MES / RTDB, FAT/SAT.